

Dear Editor,

We are pleased to submit our manuscript entitled “Machine Learning and Kalman Filtering for Precise Satellite Orbit Determination: Implications for Precise Point Positioning” for consideration as an Original Research Article in the Special Issue “Advances in GNSS Technologies for Precision Navigation” of Applied Sciences

This study investigates the impact of satellite orbit reconstruction strategies on Precise Point Positioning (PPP) performance using high-rate orbit products within a multi-GNSS framework. One-second satellite positions are reconstructed from sparse precise orbit datasets (300 s and 900 s intervals) using 9th-degree Lagrange interpolation, Extended Kalman Filter (EKF), and several machine learning algorithms, including Random Forest (RF), Support Vector Regression (SVR), K-Nearest Neighbor (KNN), and Multi-Layer Perceptron (MLP).

The proposed approaches are evaluated through a Leave-One-Out Cross-Validation (LOO-CV) framework and further assessed within the RTKLIB PPP processing environment using multi-GNSS observations. The results demonstrate that EKF and Lagrange interpolation consistently achieve superior orbit reconstruction accuracy, while machine learning models exhibit significantly larger errors under sparse sampling conditions. The PPP analysis, conducted in the East–North–Up (ENU) coordinate frame, confirms that orbit quality directly influences positioning performance, particularly in the vertical component.

The study directly contributes to the advancement of multi-GNSS technologies by addressing the challenge of generating high-rate orbit information from widely spaced precise products, which is critical for real-time and high-accuracy positioning applications. The findings emphasize the importance of orbit quality, temporal resolution, and robust estimation strategies for next-generation GNSS applications.

A key contribution of this work is the comprehensive comparison between classical interpolation, filtering-based estimation, and machine learning approaches for satellite orbit reconstruction, together with their direct impact on PPP solutions. The results highlight the current limitations of machine learning methods in high-precision GNSS applications and provide practical insights for improving multi-GNSS positioning performance.

The manuscript aligns well with the scope of Remote Sensing, particularly in the areas of multi-GNSS processing, high-precision positioning, and satellite-based Earth observation technologies. We believe that the results will be of interest to researchers and practitioners working in geodesy, satellite navigation, and advanced GNSS applications.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to Applied Sciences.

Thank you for your time and consideration. We look forward to the opportunity for our work to be reviewed.

Sincerely,

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